JAN-G-276

22 JULY 1946

JOINT ARMY-NAVY SPECIFICATION

GUM, ESTER

Army Number	Navy Number
97-54-191	52G5c

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specification:

U. S. Army	Navy Department
.....	52G5a 1 March 1941 52G5b 2 Jan. 1945

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS.

A-1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION

JAN-P-122—Packaging and Packing for Overseas Shipment—Barrels, Slack.

U. S. ARMY SPECIFICATION

100-2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

SS-R-406—Road and Paving Materials; General Specifications (Methods for Sampling and Testing).

A-2. *Other publications.*—The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[1]

INTERSTATE COMMERCE COMMISSION PUBLICATION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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B. TYPES AND CLASSES.

B–1. Ester gum covered by this specification shall be furnished in the following types and classes, as specified in the contract or order (see par. H–1):

Type I —Completely soluble, without any trace of cloud, in mineral spirits.
Class a—Low acid value.
Class b—High acid value.
Type II—Completely soluble, without any trace of cloud, in benzol.
Class c—High softening point.
Class d—Low softening point.

C. MATERIAL AND WORKMANSHIP.

C–1. See section E.

D. GENERAL REQUIREMENTS.

D–1. See section E.

E. DETAIL REQUIREMENTS.

E–1. *Chemical requirements.*—Ester gum shall conform to the requirements shown in table I.

TABLE I—*Chemical Requirements*

	Type I		Type II	
	Class a	Class b	Class c	Class d
Color	Not darker than water white	Not darker than water white	Mary to window glass	Mary to window glass
Acid number (mg KOH per gm of sample)	5 to 8	13 to 15	(8 max.)	(8 max.)
Ash, percent (max.)	0.1	0.1	0.05	0.05
Softening point, degrees F.	Not less than 203	Not less than 203	188 to 200	174 to 186

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. The material and work shall be subject at all times to inspection at the place of manufacture and every facility shall be afforded authorized Government inspectors for the prosecution of their work.

F–2. *Size of lot.*—For purposes of inspection, a lot shall not exceed 10,000 pounds and shall consist of material from not more than one manufacturing batch of ester gum. (See par. H–3.)

F–3. *Sampling.*—Three containers shall be selected, at random, from the lot for sampling purposes. If there are fewer than 3 containers in the lot, all the containers shall be sampled. A 1-pound representative sample of ester gum shall be removed, by

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F–4f. *Softening point.*—

F–4f(1). *Type I.*—

F–4f(1)a. *Apparatus (see fig. 1).*—

F–4f(1)a(1). Two circular brass plates 1/16 inch thick and 2 3/8 inches in diameter; one plate has 3 holes equally spaced about its center, each 5/8 inch in diameter. A shoulder 1/32 inch deep and 9/128 inch wide is cut around each hole so that the molds will fit snugly without danger of sliding off. These plates shall be attached to one end of a threaded steel rod, 1/8 inch by 12 inches, through their centers, by means of 3/16 inch hexagonal nuts in such manner that their opposing faces are 7/8 inch apart and parallel.

F–4f(1)a(2). Three corrosion-resisting steel molds, 3/4 inch outside diameter, 5/32 inch deep, diameter of hole at top 11/16 inch, diameter of hole at bottom 3/8 inch. Permissible variation in all dimensions of 0.01 inch.

F–4f(1)a(3). Steel ball bearings, 5/16 inch in diameter and weighing between 1.97 and 2.07 gm.

F–4f(1)a(4). Glass beaker, 1.5-liter capacity.

F–4f(1)a(5). Well annealed glass test tube 8 inches long by 2½ inches inside diameter.

F–4f(1)a(6). Two standardized thermometers 0°—400° F. range, 0.25 inch diameter, and 16 inches long.

F–4f(1)b. *Preparation of sample.*—Powder the sample in a mortar or break into pieces the size of wheat grains. Place the empty molds on an amalgamated brass plate, fill with the ground resin, apply slight heat to the brass plate, and pack the melting resin down into the mold with a spatula. Continue until the mold is filled level with the top and there are no bubbles showing. Separate the mold from the amalgamated plate and trim off the excess resin with a knife. It is convenient to make an indentation of not more than 1 mm. in the center of the molded resin to prevent the steel ball from rolling off. The resin in the molds should be a clear, fused mass. Only enough heat should be applied during filling to melt the resin sufficiently to be packed down smooth. All this operation should be done as rapidly as possible to avoid excessive loss of easily volatilized substances in the resin.

F–4f(1)c. *Procedure.*—Assemble the apparatus as shown on figure 2. See that the bottom of the inner thermometer is even with the bottom of the steel ball on top of the molded resin and the bulb of the outer thermometer is so suspended as to be opposite the lower brass plate. Fill the beaker with water-white glycerine to within ½ inch of the top when the test tube is immersed. Heat the beaker containing the glycerine, with the test tube and outer thermometer in place, to 150° F. Introduce the resin rings and inner thermometer so that the bottom plate is 1¼ inches above the bottom of the test tube. Continue heating and regulate it so there is a 4° to 5° F. per minute rise in temperature in the bath. The temperature recorded by the inner thermometer at the instant the ball drops through the ring and touches the bottom brass plate shall be taken as the softening point of the sample.

F–4f(2). *Type II.*—Determine the softening point in accordance

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means of suitable tools, from each container so chosen. Each sample shall be placed in an air-tight container and the container sealed and labeled to identify the sample with the container and lot represented. Each sample shall be subjected to all the tests specified in paragraph F–4 (see par. H–4).

F–4. *Tests.*—The samples selected in accordance with paragraph F–3 shall be subjected to test in accordance with the following paragraphs. For Navy purchases the tests shall be made at a Government laboratory unless otherwise specified in the contract or order.

F–4a. *Reagents.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and corrections shall be applied when necessary.

F–4b. *Solubility.*—Shake 50 parts, by weight, of freshly powdered ester gum in 50 parts, by weight, of mineral spirits (for type I material) or 90-percent benzol (for type II material), at 23° ± 3°C. in a suitable container. The resulting solution shall be perfectly clear.

F–4c. *Color.*—The samples shall be examined for conformance with the color requirement by comparing with known and accepted rosin standards. However, in case of dispute, United States rosin standards, or approved duplicates thereof, prepared and maintained by the United States Department of Agriculture, Washington, D. C., shall be used.

F–4d. *Acid number.*—Weigh approximately 3.000 gm. of freshly powdered ester gum and transfer to a 500-ml. Erlenmeyer flask. Add 200 ml. of a mixture of equal parts, by weight, of 95-percent ethyl alcohol and benzol. (This mixture should be titrated previously to a very faint pink with standard alkali solution, using phenolphthalein as indicator.) When solution is obtained, add phenolphthalein indicator and titrate to a faint permanent pink color with 0.1 N KOH solution. Calculate the acid number as follows:

$$\text{Acid number} = \frac{56.1\ AN}{W}$$

where

A = ml. standard KOH used in titration
N = normality of KOH solution
W = weight of sample.

F–4e. *Ash.*—Weigh approximately 10.00 gm. of the sample in a platinum dish, ignite to constant weight in a muffle furnace, cool in a desiccator, and weigh. Calculate the percentage of ash as follows:

$$\text{Percentage of ash} = \frac{100A}{W}$$

where

A = weight of residue
W = weight of sample.

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with method 22 prescribed in Federal Specification SS–R–406, with the exception that the ring shall be a brass, shouldered ring conforming to the following dimensions: The overall depth shall be ¼ inch. The outside diameter shall be 29/32 inch for a depth of 11/64 inch from the top surface and ¾ inch for the balance of the ¼-inch depth. The inside diameter shall be 25/32 inch for a depth of 9/64 inch from the top surface and ⅝ inch for the remaining 7/64 inch. The inner shoulder shall have a rounded fillet.

F–5. *Rejection and resubmission.*—If any one of the samples does not conform to any of the specified requirements, the entire lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on each container in the lot at no expense to the Government. The contractor may then remove defective portions of the lot and resubmit the lot for acceptance.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packing (for domestic or overseas shipment).*—

G–1a. *Type I.*—Unless otherwise specified in the contract or order, type I ester gum shall be packed in removable head, steel drums conforming to the requirements of Interstate Commerce Commission Specification 37D or in paper-lined, No. 1 slack barrels conforming to the requirements of Joint Army-Navy Specification JAN–P–122 (see par. H–1). A single container, when packed for shipment, shall contain not more than 250 pounds net.

G–1b. *Type II.*—Unless otherwise specified in the contract or order, type II ester gum shall be packed, as a solid mass of approximately 400 pounds net weight, in removable-head steel drums conforming to the requirements of Interstate Commerce Commission Specification 37D.

G–2. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should contain the following features:

(a) Title, number, and date of the specification.
(b) Type and class of ester gum required (see par. B–1).
(c) Kind of shipping container required for type I ester gum (see par. G–1a).

H–2. *Uses.*—Type I ester gum is intended for use in the manufacture of waterproof spar varnish. Type II ester gum is intended for use in the manufacture of shoe impregnite.

H–3. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–4. If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion and with the concurrence of the contractor, may modify in whole on in part

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the sampling and testing procedures specified in this specification. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures prescribed in this specification.

H–5. This specification replaces Chemical Warfare Service Tentative Specification 197–54–191, dated 3 June 1942.

H–6. Copies of Interstate Commerce Commission "Regulations for the Transportation of Explosives and Other Dangerous Articles," of which I.C.C. Specification 37D forms a part, may be obtained upon application, accompanied by postal note (or money order), coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 40 cents.

H–7. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy and Federal Specifications Used by the War Department." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

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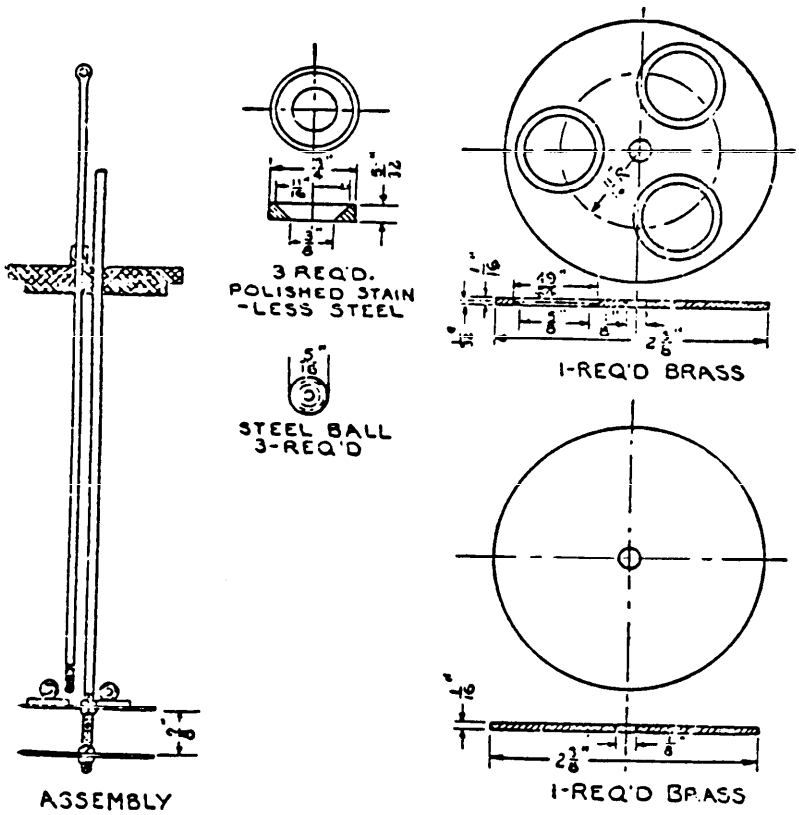


FIGURE 1.

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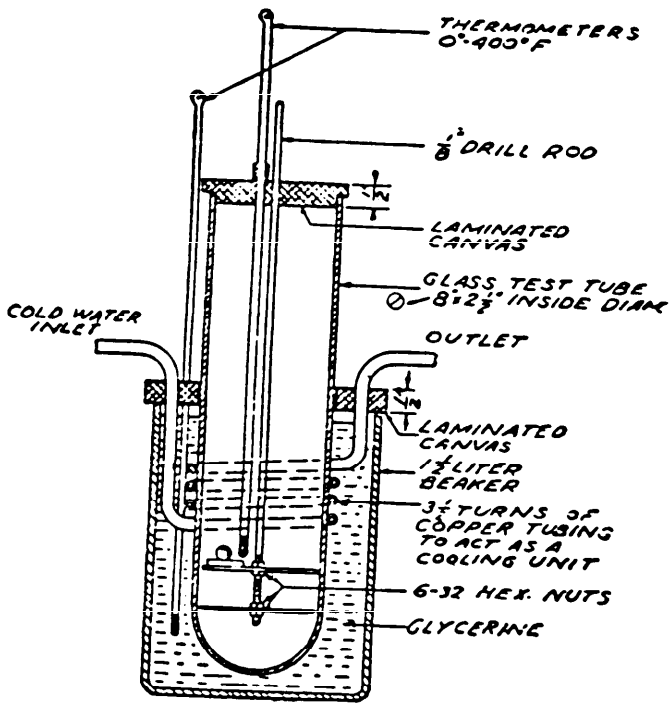


FIGURE 2.

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